Vaishnavi Sawant

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EDUCTATION

Masters of Technology - Electronic Design and Technology **2023 - 2025**
- National Institute of Technology, Calicut 8.84 CGPA

Bachelor of Technology - Electronics and Telecommunication Engineering **2017 – 2021**
- Government College of Engineering Karad, Maharashtra 8.60 CGPA

EXPERIENCE

Co/op Intern | AMD, Bengaluru (*January – July 2025*)

- Integrated multiple IPs for **NBIO–IOHUBSS client design**, ensuring functional and architectural correctness.
- Executed design quality checks including connectivity, lint, CDC, and synthesis.
- Built automation scripts (Python, Bash, TCL) to
  - Extract tool configurations and shared components from sources files, eliminating manual updates.
  - Implement credit alignment and synchronization across multiple IP blocks, enabling seamless flow control and preventing buffer overflows in credit-based protocols.

SKILLS

Technical Skills: Verilog, System Verilog, Digital Design, Logic Design and Verification, Design for Testability (DFT), C Programming, FPGA, Static timing analysis (STA), Linux, CDC analysis, Communication Protocols (SPI, I2C, UART), Computer Architecture, IP Design, Python (Basic), Perl, TCL.

Tools: Xilinx Vivado, Cadence virtuoso, Cadence Genus, Vitis, Verdi.

Subjects: Digital Electronics, VLSI Architecture, Digital IC Design, FPGA, Hardware Software Codesign.

PUBLICATIONS

Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition

Presented paper at VLSI Design and Test (VDAT) 2025 Conference: "Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition" – introduced two novel architectures based on the Hestenes-Jacobi algorithm, achieving up to 90% area and power reduction in Cadence Genus (GPDK 90nm @100 MHz) (Accepted for publication).

PROJECTS

Implementation Of Hardware-Efficient Matrix Multiplication using Tiled Singular Value Decomposition *(May – Dec 2024)*

- Designed and implemented two novel hardware architectures using Tiled Singular Value Decomposition (TSVD) to accelerate matrix multiplication for deep neural network (DNN) workloads.
- Achieved major hardware optimizations: • Hestenes-Tile Fusion Architecture: ~75% area, 77% power, and 69% gate count reduction. • Hybrid SVD-Tiled Architecture: ~90% reduction in area, power, and gates, with improved latency and throughput.
- Validated and synthesized designs in Cadence Genus (GPDK 90nm, 100 MHz), ensuring functionality and performance efficiency.

Design of I2C Communication protocol using Verilog and Verification using System Verilog *(May – June 2024)*

- Developed I2C communication protocol in Verilog and verified its operation using System Verilog, ensuring proper functionality and effective communication between master and multiple slave devices, and enhancing system reliability
- Gained in-depth knowledge of the protocol while demonstrating good skills in digital design and verification using System Verilog testbenches.

Implementation of Asynchronous and Synchronous FIFO in Verilog and Verification of Synchronous FIFO using System Verilog *(March – April 2024)*

- Designed a robust Asynchronous and Synchronous FIFO memory in Verilog HDL using AMD Xilinx Vivado to enable seamless data transfer between two systems, effectively handling Full and Empty conditions to prevent overflow and underflow, ensuring reliable functionality and optimal performance.
- Successfully verified the operation of a Synchronous FIFO using System Verilog, ensuring proper functionality of the FIFO and demonstrating a good understanding of the System Verilog testbench environment.

LANGUAGES

English, Hindi & Marathi

ADITIONAL INFORMATION

Awarded the prestigious Nurturing Brilliance Cummins Scholarship (2018-2021) by Cummins India, recognizing meritorious students across engineering disciplines.